

BAMSurgeon: Methods for spike-in mutations on BAM files

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1 Introduction

1.1 Software Dependencies

BAMSurgeon requires the following software packages be available:

1. samtools/wgsim/tabix (<http://samtools.sourceforge.net/>)
2. pysam (<http://code.google.com/p/pysam/> or `pip install pysam`)
3. bwa (<http://bio-bwa.sourceforge.net/>)

For install instructions please see the included README.md or have a look at the Dockerfile.

1.2 Known bugs and limitations

BAMSurgeon is under continuous development driven by suggestions and bug reports from the mutation calling community.

1. If haplotype resolution is important for your application we suggest phasing BAMs beforehand and simulating mutations on a per-haplotype basis.

2 Adding SNVs to existing BAM alignments (addsnv)

2.1 Usage

```
usage: addsnv.py [-h] -v VARFILENAME -f BAMFILENAME -r REFFASTA -o OUTBAMFILE
               [-s SNVFRAC] [-m MUTFRAC] [-n NUMSNVS] [-c CNVFILE]
               [-d COVERDIFF] [-z HAPLOSIZE] [-p PROCS]
               [--mindepth MINDEPTH] [--maxdepth MAXDEPTH]
               [--minmutreads MINMUTREADS] [--avoidreads AVOIDREADS]
               [--nomut] [--ignoresnps] [--ignoreref] [--force] [--insane]
               [--single] [--maxopen MAXOPEN] [--requirepaired] [--tagreads]
               [--skipmerge] [--ignorepileup] [--aligner ALIGNER]
               [--alignerthreads ALIGNERTHREADS] [--alignopts ALIGNOPTS]
               [--tmpdir TMPDIR] [--seed SEED] [--vcf VCF]
```

adds SNVs to reads, outputs modified reads as .bam along with mates

options:

- h, --help show this help message and exit
- v VARFILENAME, --varfile VARFILENAME
 Target regions to try and add a SNV, as BED
- f BAMFILENAME, --bamfile BAMFILENAME
 sam/bam file from which to obtain reads
- r REFFASTA, --reference REFFASTA
 reference genome, fasta indexed with bwa index _and_
 samtools faidx
- o OUTBAMFILE, --outbam OUTBAMFILE
 .bam file name for output
- s SNVFRAC, --snvfrac SNVFRAC
 maximum allowable linked SNP MAF (for avoiding
 haplotypes) (default = 1)
- m MUTFRAC, --mutfrac MUTFRAC
 allelic fraction at which to make SNVs (default = 0.5)
- n NUMSNVS, --numsnvs NUMSNVS
 maximum number of mutations to try (default: entire
 input)
- c CNVFILE, --cnvfile CNVFILE
 tabix-indexed list of genome-wide absolute copy number
 values (e.g. 2 alleles = no change)
- d COVERDIFF, --coverdiff COVERDIFF
 allow difference in input and output coverage
 (default=0.9)
- z HAPLOSIZE, --haplosize HAPLOSIZE
 haplotype size (default = 0)
- p PROCS, --procs PROCS
 split into multiple processes (default=1)
- mindepth MINDEPTH minimum read depth to make mutation (default = 10)
- maxdepth MAXDEPTH maximum read depth to make mutation (default = 2000)
- minmutreads MINMUTREADS
 minimum number of mutated reads to output per site
- avoidreads AVOIDREADS
 file of read names to avoid (mutations will be skipped
 if overlap)
- nomut dry run
- ignoresnps make mutations even if there are non-reference alleles
 sharing the relevant reads
- ignoreref make mutations even if the mutation is back to the
 reference allele
- force force mutation to happen regardless of nearby SNP or
 low coverage
- insane ignore sanity check enforcing input read count =
 output read count in realignment
- single input BAM is single-ended (default is paired-end)
- maxopen MAXOPEN maximum number of open files during merge (default
 1000)

```

--requirepaired      skip mutations if unpaired reads are present
--tagreads           add BS tag to altered reads
--skipmerge          final output is tmp file to be merged
--ignorepileup       do not check pileup depth in mutation regions
--aligner ALIGNER     supported aligners: STAR,backtrack,bowtie2,bwakit,gsna
                     p,mem,minimap2,novoalign,tmap
--alignerthreads ALIGNERTHREADS
                     threads used per realignment (default = 1)
--alignopts ALIGNOPTS
                     aligner-specific options as comma delimited list of
                     option1:value1,option2:value2,...
--tmpdir TMPDIR       temporary directory (default=addsnv.tmp)
--seed SEED           seed random number generation
--vcf VCF             Path for the output VCF file. If not provided, the
                     file will be saved in the current directory.

```

2.2 Description

Single nucleotide changes are introduced to an existing BAM alignment using `addsnv.py` as outlined in Figure 1. Input consists of a list of locations where SNVs will be made, a target BAM alignment, and a reference genome indexed using `bwa` (preferably the same genome used to align the reads in the BAM file).

2.3 Input

Mutation input (-v/-varfile) Input to `addsnv.py` (-v/-varfile) is a three column format plus an extra optional column for the desired variant allele fraction (VAF). Input coordinates are **1-based**, meaning they will match between input to -v/-varfile, VCF, and `samtools mpileup`. Note that `addindel.py` uses a 1-based coordinate scheme. The default VAF is specified via -m/-mutfrac. For example:

```

22 34166720 34166720 0.25
22 33908770 33908770 0.15
22 33714964 33714964 0.5
22 33769483 33769483 0.75
22 33958087 33958087 0.01
22 34264774 34264774 0.25
22 33702084 33702084 0.2

```

Specific bases can also be specified using a fifth input column:

```

22 34166720 34166720 0.25 A
22 33908770 33908770 0.15 T
22 33714964 33714964 0.5 G
22 33769483 33769483 0.75 C

```

The above input will cause `addsnv` to attempt to add SNVs at the specified locations, using the VAF specified in the fourth column if available, otherwise defaulting to 0.5 or whatever is specified

by `-m/-mutfrac` (e.g. the last three lines). If a position is specified that spans more than one base, the position of the SNV input will be chosen at random from that interval.

BAM Alignment input (`-f/-bamfile`) BAMs ideally should adhere to SAM specification i.e. they should validate via `picards ValidateSamFile`. BAMs should be sorted in coordinate order and indexed. BAMs should consist either of entirely paired-end reads or entirely single-end (fragment) reads using the `-single` option, mixing paired and single reads yields unpredictable results. Currently, `bwa backtrack` (`bwa aln`) and `bwa mem` are supported, although it should be straightforward enough to add further aligners through modification of the `remap_paired` function. In practice we have been able to spike mutations into RNA-seq MapSplice BAMs, for instance.

Reference genome (`-r/-reference`) The reference genome should be the **same as that used to create the target BAM file**, specifically the chromosome names and lengths in the reference FASTA must be the same as in the BAM header. The reference must be indexed for `bwa` (`bwa index`) and indexed with `samtools` (`samtools faidx`).

Avoiding existing SNPs with `-s/-snpfrac` (optional, default=1.0) In an attempt to preserve haplotype structure and avoid confusion by spiking mutations on top of existing heterozygous alleles, the `-s/-snpfrac` cutoff may be useful. For a selected mutation site S and specified cutoff f starting from the leftmost base L of the leftmost read containing S and ending at the rightmost base R of the rightmost read containing S , the minor allele frequency M for every intervening base is calculated for every base in the interval $[L, R]$. If $M < f$ for all bases in $[L, R]$ the mutation at S is allowed, and skipped otherwise. Note, do not confuse this with `-m/-mutfrac`, which is the target variant allele frequency for mutations unless overridden in the mutation input file. By default this function is turned off (threshold minor allele frequency set to 1.0). Setting this too low can yield few successful mutations as sequence errors will inhibit successful spike-in, we recommend a setting of 0.1 for Illumina GA/HiSeq reads.

CNV file (`-c/-cnvfile`) (optional) It is possible to simulate adding mutations to a single haplotype for copy-number variant regions. For example, if a region has absolute copy number 3 (instead of 2), a mutation into one haplotype should have an apparent VAF of roughly 0.33 instead of 0.5. This optional file should be in BED-3 format plus a fourth column representing the copy number fold-change for the region defined by the chromosome/contig, start, and end fields. A copy number fold-change of 1 will not adjust the variant allele fractions. Copy number changes below 1 will adjust VAFs higher and copy number fold-changes above 1 will adjust VAFs correspondingly lower.

Additional options

- `-d/-coverdiff` : threshold on output coverage divided by input coverage for mutation sites. Mutations can cause alignments to change, if this is undesired set this high, if this is part of the fun, set this low.
- `-z/-haplosize` : specifies size of haplotype block to consider when adding more than 1 proximal mutation. For instance, if two SNVs are spiked in 10bp apart and `-haplosize` is 10 or greater, the two SNVs will be on the same haplotype (i.e. share the same reads for reads covering both positions)

- `-p/-procs` : BAMSurgeon programs currently spawn a new process for each spikein, to speed things along the processes can be run concurrently with the number of simultaneous jobs set by this parameter.
- `-mindepth` : minimum depth at sites for mutations, sites with fewer reads in pileup column will be dropped. Default is 10 reads.
- `-maxdepth` : sites with greater than this number of covering reads will be dropped. Default is 2000 reads. (May need to increase for targeted sequencing data) The cutoff is raised in proportion to local copy number, taken from `--cnvfile` and, in a unified `bamsurgeon` run, from any duplication spiked into the same locus earlier in that run — amplified sequence carries more reads legitimately, and would otherwise be dropped for it.
- `-minmutreads` : minimum number of mutated reads per site. If this level of coverage cannot be obtained, mutations will be dropped.
- `-ignoresnps`: ignores existing non-reference bases at the mutation site, required if the goal is to change a non-reference allele back to reference
- `-ignoreref`: required if the goal is to change a non-reference allele back to reference
- `-avoidreads` : `addsnv` and `addindel` can be passed a list of read names to avoid when making mutations. Any sites selected that overlap these reads will be dropped. This feature exists as a means to ensure mutations across multiple runs do not overlap.
- `-force` : force the addition of a mutation even if it fails thresholds set by `-mindepth` and `-c/-coverdiff`. Not recommended for most use cases.
- `-insane`: ignores sanity checking number of input reads versus number of output reads. This seems to be required for some aligners or situations where unaligned reads are dropped or secondary alignments are added.
- `-single` : covered above, assume reads are single-end (i.e. fragment reads).
- `-maxopen` : set the maximum number of open files (check your system limit with `ulimit -a`)
- `-requirepaired` : drop mutations where a paired end in the region cannot be located. However, it is probably better to correct the problem in the BAM file than to resort to using this option.
- `-skipmerge` : instead of replacing the mutated and realigned reads back into the original BAM, which is normally the final step, leave the `addsnv`.
- `-aligner` : choose one of the supported aligners, default is `backtrack` (for `bwa backtrack`). Additional aligners can be added through modification of `bs/aligners.py`, and I try to accommodate requests for widely-used aligners if they are not already implemented. Aligner specific options are handled through `-alignopts`
- `-alignopts` : aligner-specific options are specified in a comma-delimited list as `option1:value1,option2:value2,etc`. Required options are checked by the the function `checkoptions` in `bs/aligners.py`. For example, if `-aligner novoalign` is specified, `-alignopts novoref:path to novoalign reference;` should also be specified.
- `-tagreads` : adds a 'BS' tag to the reads altered by `bamsurgeon`, useful in debugging

- `--seed` : sets a random seed for reproducibility
- options not listed here are either new and the manual has not been updated as yet, or they are there for debugging and will be removed in the future.

3 Adding INDELs to existing BAM alignments (addindel)

3.1 Usage

```
usage: addindel.py [-h] -v VARFILENAME -f BAMFILENAME -r REFFASTA -o
  OUTBAMFILE [-s SNVFRAC] [-m MUTFRAC] [-n NUMSNVS]
  [-c CNVFILE] [-d COVERDIFF] [-p PROCS]
  [--mindepth MINDEPTH] [--maxdepth MAXDEPTH]
  [--minmutreads MINMUTREADS] [--avoidreads AVOIDREADS]
  [--nomut] [--det] [--force] [--insane] [--single]
  [--maxopen MAXOPEN] [--requirepaired] [--aligner ALIGNER]
  [--alignopts ALIGNOPTS] [--alignerthreads ALIGNERTHREADS]
  [--tagreads] [--skipmerge] [--ignorepileup]
  [--tmpdir TMPDIR] [--seed SEED] [--vcf VCF]
```

adds INDELs to reads, outputs modified reads as .bam along with mates

options:

```
-h, --help          show this help message and exit
-v VARFILENAME, --varfile VARFILENAME
    Target regions to try and add a SNV, as BED
-f BAMFILENAME, --bamfile BAMFILENAME
    sam/bam file from which to obtain reads
-r REFFASTA, --reference REFFASTA
    reference genome, fasta indexed with bwa index _and_
    samtools faidx
-o OUTBAMFILE, --outbam OUTBAMFILE
    .bam file name for output
-s SNVFRAC, --snvfrac SNVFRAC
    maximum allowable linked SNP MAF (for avoiding
    haplotypes) (default = 1)
-m MUTFRAC, --mutfrac MUTFRAC
    allelic fraction at which to make SNVs (default = 0.5)
-n NUMSNVS, --numsnvs NUMSNVS
    maximum number of mutations to try (default: entire
    input)
-c CNVFILE, --cnvfile CNVFILE
    tabix-indexed list of genome-wide absolute copy number
    values (e.g. 2 alleles = no change)
-d COVERDIFF, --coverdiff COVERDIFF
    allow difference in input and output coverage
    (default=0.1)
-p PROCS, --procs PROCS
```

```

        split into multiple processes (default=1)
--mindepth MINDEPTH    minimum read depth to make mutation (default = 10)
--maxdepth MAXDEPTH    maximum read depth to make mutation (default = 2000)
--minmutreads MINMUTREADS
        minimum number of mutated reads to output per site
--avoidreads AVOIDREADS
        file of read names to avoid (mutations will be skipped
        if overlap)
--nomut                dry run
--det                  deterministic base changes: make transitions only
--force                force mutation to happen regardless of nearby SNP or
        low coverage
--insane                ignore sanity check enforcing input read count =
        output read count in realignment
--single                input BAM is single-ended (default is paired-end)
--maxopen MAXOPEN      maximum number of open files during merge (default
        1000)
--requirepaired        skip mutations if unpaired reads are present
--aligner ALIGNER      supported aligners: STAR,backtrack,bowtie2,bwakit,gsna
        p,mem,minimap2,novoalign,tmap
--alignopts ALIGNOPTS
        aligner-specific options as comma delimited list of
        option1:value1,option2:value2,...
--alignerthreads ALIGNERTHREADS
        threads used per realignment (default = 1)
--tagreads              add BS tag to altered reads
--skipmerge             final output is tmp file to be merged
--ignorepileup          do not check pileup depth in mutation regions
--tmpdir TMPDIR         temporary directory (default=addindel.tmp)
--seed SEED             seed random number generation
--vcf VCF               Path for the output VCF file. If not provided, the
        file will be saved in the current directory.

```

3.2 Description

Short insertions and deletions (commonly referred to as INDELs) can be added to BAM files using addindel.py. The options are essentially the same as for addsnv.py, but the syntax of the input file is slightly different and, importantly is **0-based** (i.e. constant with BED files):

```

22 33694303 33694304 0.50 INS ATGGC
22 33815273 33815274 0.25 INS CG
22 33920490 33920491 0.33 INS ATTTGCTTAGCTGAGGGCTTAGGCTTAGCATGCGTA
22 33944542 33944543 0.50 DEL
22 34006643 34006653 0.50 DEL
22 33958087 33958090 0.40 DEL

```

The first four columns are the same as for addsnv.py input but the fourth column (VAF) is mandatory. The fifth column must be either 'INS' or 'DEL' (for insertion or deletion). For insertions

there must be a sixth column specifying the sequence to be inserted at the position indicated in the second column (the position in the third column does not matter for insertions). In the above example, the first three columns direct addindel.py to (1) add an insertion of “ATGGC” at chromosome 22 position 33694303 with 50% VAF, (2) add an insertion of “CG” at position 338152273 with 25% VAF, and (3) add an insertion of “ATTTGC...” with 33% VAF.

For deletions, the sequence from the start to the end (columns two and three) is deleted. Lines 4,5, and 6 above indicate deletions of 1 bp, 10bp, and 3 bp, respectively.

4 Adding SVs to existing BAM alignments (addsv.py)

4.1 Usage

```
usage: addsv.py [-h] -v VARFILENAME -f BAMFILENAME -r REFFASTA -o OUTBAMFILE
  [-l MAXLIBSIZE] [-s SVFRAC] [--pad PAD] [--readlen READLEN]
  [--mindepth MINDEPTH] [--maxdepth MAXDEPTH]
  [--maxdfraction MAXDFRAC] [-n MAXMUTS]
  [-c CNVFILE] [--donorbam DONORBAM] [--ismean ISMEAN]
  [--issd ISSD] [--simerr SIMERR] [-p PROCS] [--inslib INSLIB]
  [--aligner ALIGNER] [--alignopts ALIGNOPTS]
  [--alignerthreads ALIGNERTHREADS] [--tagreads] [--skipmerge]
  [--keepsecondary] [--debug] [--tmpdir TMPDIR] [--seed SEED]
  [--allowN] [--vcf VCF]
```

adds SVs to reads, outputs modified reads as .bam along with mates

options:

```
-h, --help          show this help message and exit
-v VARFILENAME, --varfile VARFILENAME
                    whitespace-delimited target regions for SV spike-in,
                    see manual for syntax
-f BAMFILENAME, --bamfile BAMFILENAME
                    sam/bam file from which to obtain reads
-r REFFASTA, --reference REFFASTA
                    reference genome, fasta indexed with bwa index _and_
                    samtools faidx
-o OUTBAMFILE, --outbam OUTBAMFILE
                    .bam file name for output
-l MAXLIBSIZE, --maxlibsize MAXLIBSIZE
                    maximum fragment length of seq. library
-s SVFRAC, --svfrac SVFRAC
                    allele fraction of variant (default = 1.0)
--pad PAD           reference flank each side of a breakpoint
                    (default = 2x --maxlibsize)
--readlen READLEN   simulated read length (default: modal read length
                    sampled from the BAM)
--mindepth MINDEPTH minimum read depth in the breakend position to make
                    mutation (default = 10)
--maxdepth MAXDEPTH maximum read depth in the breakend position to make
```



```

        mutation (default = 2000)
--maxdfrac MAXDFRAC    maximum discordant fraction (is_proper_pair / is_pair)
                        of reads (default = 0.1)
-n MAXMUTS             maximum number of mutations to make
-c CNVFILE, --cnvfile CNVFILE
                        tabix-indexed list of genome-wide absolute copy number
                        values (e.g. 2 alleles = no change)
--donorbam DONORBAM    bam file supplying duplication interior reads;
                        without it the interior is simulated
--ismean ISMEAN        mean insert size (default = estimate from region)
--issd ISSD            insert size standard deviation (default = estimate
                        from region)
--simerr SIMERR        error rate for wgsim-generated reads
-p PROCS, --procs PROCS
                        split into multiple processes (default=1)
--inslib INSLIB        FASTA file containing library of possible insertions,
                        use INS RND instead of INS filename to pick one
--aligner ALIGNER      supported aligners: STAR,backtrack,bowtie2,bwakit,gsna
                        p,mem,minimap2,novoalign,tmap
--alignopts ALIGNOPTS
                        aligner-specific options as comma delimited list of
                        option1:value1,option2:value2,...
--alignerthreads ALIGNERTHREADS
                        threads used per realignment (default = 1)
--tagreads             add BS tag to altered reads
--skipmerge            do not merge spike-in reads back into original BAM
--keepsecondary        keep secondary reads in final BAM
--debug               output read tracking info to debug file, retain all
                        intermediates
--tmpdir TMPDIR        temporary directory (default=addsv.tmp)
--seed SEED            seed random number generation
--allowN               allow N in the reference slice, replace with A and warn
                        (default: drop mutation)
--vcf VCF              Path for the output VCF file. If not provided, the
                        file will be saved in the current directory.

```

4.2 Description

Larger structural variants (insertions, deletions, duplications, inversions, and compound rearrangements) are added to existing BAM alignments using `addsv.py` as described in Figure 3. Input consists of a list of regions where SVs will be made along with a specification of each variant, and as with `addsnv` and `addindel`, a target BAM alignment, and a reference genome indexed using `bwa` are required.

4.3 Input

The input mutation list consists of four columns: chromosome, start of region, end of region, and a controlled-vocabulary description of the mutation. The mutated haplotype is built from reference sequence spanning the region plus `--pad` bases either side (`--pad` defaults to twice `-l/--maxlibsize`), so that simulated read pairs crossing each junction are fully contained. The mutation description starts with either `INS`, `DEL`, `DUP`, `TRN`, or `INV` for insertion, deletion, duplication, transduction, and inversion, respectively and is followed by mutation-specific options. An example follows.

```
22 34171055 34184700 INS ACTAACCTGCACAATGTGCACATGTACCCTAAAACTTAGAGTATAATAAAAAAAAA 13 AA^TTTT 0.9
22 33607508 33607508 TRN 22 34314981 34314981 ++ 1.0
22 33871043 33874754 DEL 0.75
22 33891043 33902689 BIGDEL 0.5
22 34071043 34084754 INV 0.5
22 34271043 34284754 DUP 3 0.9
```

For translocations (`TRN`), the above example will join the 6kbp region centered on 22:33607508 and the 6kbp region centered on 22:34314981. The variant allele fraction (VAF) is specified in the last column (0.9 in the above example).

For insertions, `INS` should be followed by either a FASTA file containing the sequence to be inserted, by the nucleotide sequence itself, or by `RND` if a library of potential insertion sequences is passed in FASTA format via the `-inslib` option. For example, `INS ATG` would insert the sequence ‘ATG’ at the midpoint of the specified region, while `INS LINE1.fa` would insert the sequence in the FASTA-formatted file `LINE1.fa` there. The VCF records the position the insertion actually landed at, which is the midpoint unless a cut-site motif moved it. Many insertions include target site duplications (TSDs), so to simulate this an integer value can be added in the fifth column of insertion entries specifying the TSD length. Endonuclease motifs are another common feature of insertion breakpoints, so to simulate this, a preferred cut site can be defined in the 6th column of an insertion entry with the syntax `NNN^NNN` where ‘NNN’ denotes sequence of any length composed of characters (A, T, G, C) with the actual cut site in the motif specified by the caret (^). An optional seventh column can be used to specify the VAF, 0.8 in the example above.

For deletions `DEL` is optionally followed by specifying the per-mutation VAF (0.75 in the example above). Inversions (`INV`) have no further options aside from an optional VAF column - the specified region will be inverted. Duplications (`DUP`) have two optional parameters, an integer specifying the number of additional tandem copies, e.g. `DUP 2` adds two extra copies, and a VAF column. In general, we recommend that short insertions and deletions (those shorter than one read length) be created with `addindel.py` instead of `addsv.py`, as `addindel.py` edits existing reads rather than simulating replacements.

Compound variants chained together in a semicolon-delimited list, e.g. `DUP 1;DEL 0.5;INS AAATCC;INV`, are not currently supported: such lines are reported and skipped rather than being applied partially.

4.4 Mutation Size

Mutation types `DEL`, `DUP`, and `INV` are handled identically regardless of size: the mutated haplotype is built from reference sequence spanning the interval plus `--pad` bases either side, and reads are simulated over it. The names `BIGDEL`, `BIGDUP`, and `BIGINV` are accepted as deprecated spellings of `DEL`, `DUP`, and `INV` and are treated identically; there is no longer a size threshold

that switches between implementations. For duplications, the interior copy is simulated by default. Supplying `--donorbam` instead takes the interior reads from that BAM, which keeps the simulated template small for large duplications and gives the interior a real error profile. Ideally the donor BAM and the input BAM would be separate partitions of the same higher coverage BAM.

4.5 Exact Breakpoints

Breakpoints are created at the requested positions. Because the mutated haplotype is built from reference sequence rather than from a local assembly, the coordinates in the output VCF are exact by construction and match the input.

4.6 Controlling SV VAF

Variant allele fraction for `addsv.py` can be controlled either via the `-c/--cnvfile` option (see Section 2.3) or through addition of a column in the input file (float value between 0.0 and 1.0). Note that all other optional columns must be specified for the VAF column to be effective, which means the input format for `addsv.py` really needs an overhaul.

4.7 Output

Where mutations add sequence (e.g. duplications), new reads are created that will be added to the original BAM, and where mutations remove sequence (e.g. deletions) those reads are marked as excluded (excluded read names are recorded in the file specified by `-x/--excluded`). Excluded reads are not carried over from the original BAM to the mutated BAM, creating the copy number effect associated with deletion.

4.8 Additional Options

Here are some extended descriptions for additional options not mentioned yet:

- `-s/-svfrac` : equivalent to `-m/-mutfrac` for `addindel.py` and `addsnv.py`
- `-n` : only attempt this many mutations
- `-ismean/-issd` : the mean and standard deviation of the library size should be estimated from the original bam and specified through these arguments. This can be accomplished a number of ways; `samtools stats` reports both as `insert size average` and `insert size standard deviation`.
- `-simerr` : control sequencing error rate around breakends
- `-inslib` : a FASTA file can be specified and insertions can be tagged with RND instead of a sequence or file. If this is done, a sequence will be chosen at random from the FASTA file specified by `-inslib` and inserted.
- `-skipmerge` : same meaning as for `addsnv/addindel`
- `-allowN` : useful if desired mutations are being dropped due to one or more N bases present in the reference sequence of the region

5 Output (all tools)

Output consists of a BAM file containing the spike-in mutations, a VCF file containing the results of the spike-in including expected variant positions and VAFs, and a directory of log files describing which bases were changed in which reads (one log file per mutation). For SVs this file contains the local assemblies for each region where a mutation was made. If needed, the directory containing output logs can be transformed into VCF format using the script `makevcf.py`, `makevcf_indel.py`, or `makevcf_sv.py` as appropriate, located the `scripts/` subdirectory.

bamsurgeon_snv-eps-converted-to.pdf

Figure 1: Method for adding SNVs to existing BAM alignments. Starting with the original BAM alignment (a) and a list of positions (b), reads overlapping the chosen positions are selected the target base change is made (c). Altered reads are re-mapped the the reference genome using bwa, and the modified, realigned reads replace the unmodified versions in the original BAM (d).



bamsurgeon_sv_del-eps-converted-to.pdf

Figure 2: Method for adding multi-nucleotide variants (indels and SVs) to existing BAMs: deletion example. Starting with the original BAM alignment (a) the mutated haplotype for a region, e.g. (b), is built directly from reference sequence and the desired mutation(s) are created in it. Read coverage is simulated over this haplotype using wgsim (<https://github.com/lh3/wgsim>, also included with samtools) with parameters `-e 0 -r 0 -R 0` to suppress additional mutations, other parameters are set based on the input BAM and desired coverage. Simulated coverage is scaled based on whether the mutation is adds sequence (duplication) or removes sequence (deletions), and replaced into the original BAM while excluded reads are removed from the original BAM (d).

6 Testing

The `test/` directory contains a pytest suite. Run it from the repository root:

```
python3 -m pytest
```

`test/cases.py` holds the matrix of spike-in configurations and `test/test_spikein.py` runs each one, then checks the output BAM against the truth VCF that run emitted using `scripts/validate_spikein.py`. Each case gets its own temporary directory, so cases do not interfere with one another.

Cases naming an aligner that is not installed are skipped rather than failed. A run reporting only skips is therefore not evidence of anything: check that the passed count is what you expect.

`test/test_evaluator.py` covers `scripts/evaluator.py` and needs only pysam, so it runs anywhere. `test/test_combined.py` exercises the unified `bamsurgeon` entry point on a VCF containing SNVs, indels and SVs together.

7 Other Considerations

7.1 System Requirements

For each process (specified by `-p/--procs`) `add*.py` will require 4GB RAM. So for 8 concurrent mutation processes, you should have 32GB RAM. System requirements may decrease in the future as we find ways to improve efficiency. For more recent versions of bwa, it is possible to load the reference index into shared memory with `'bwa shm'` which seems to improve performance.

7.2 Alignments

Reads should be re-aligned using the same aligner and parameters as were used to create the original BAM file. By default, `bamsurgeon` assumes alignments were performed using `bwa aln`. Specific parameters that are not currently exposed on the command line may be altered through modification of the functions `remap_paired` (for paired reads) or `remap_single`. Supported aligners currently include: `bwa aln`, `bwa mem`, `novalign`, `gsnap`, `STAR`, `bowtie2`, `tmap`, `bwakit`.

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